

Problem Set 1

Probability Theory for Computer Science - Week 1

☺ Basic ! Important ★ Challenging (with hints)

☺ Basic level

- ☺ 1. *Binary classifier outcomes.* An email has a true label in $\{S, L\}$, where S means spam and L means legitimate. An AI classifier also predicts S or L. Represent an outcome as an ordered pair (true label, predicted label).
- (a) Write the sample space Ω .
 - (b) Define C, the event that the classifier is correct.
 - (c) Define F, the event of a false positive: a legitimate email is classified as spam.
 - (d) Write C^c .
- ☺ 2. *Structured output from a language model.* A language-model response is placed in exactly one category: V = valid JSON with the required schema, W = valid JSON with the wrong schema, or I = invalid JSON.
- (a) Write a sample space.
 - (b) Define P, the event that the response can be parsed as JSON.
 - (c) Define U, the event that the response is immediately usable.
 - (d) Find P^c and $P \setminus U$.
- ☺ 3. *Randomized search node.* A randomized search algorithm selects one of eight nodes, numbered 1 through 8, with equal probability. Let A be the event that an even-numbered node is selected and B the event that a node in the first half is selected.
- (a) List A, B, $A \cap B$, and $A \cup B$.
 - (b) Compute the probability of each of these four events.
- ☺ 4. *Compute-server state.* At a randomly observed moment, an AI compute server is idle, busy, or failed. The proposed probabilities are 0.20, 0.70, and 0.10, respectively.
- (a) Verify that this is a valid probability model.
 - (b) Let A be the event that the server is available for work, meaning idle or busy. Find $P(A)$.
 - (c) Let B be the event that the server is not idle. Find $P(B)$.

- ☺ 5. *Two model evaluations.* Two AI model evaluations are run. Each evaluation is recorded as pass (P) or fail (F). Do not assume any probabilities.
- Write the sample space using ordered pairs.
 - Define E: exactly one evaluation fails.
 - Define A: at least one evaluation fails.
 - Express 'both evaluations pass' as a complement of A.
- ☺ 6. *A tiny token vocabulary.* The next token is selected uniformly from $\Omega = \{\text{if, else, for, while, return}\}$. Let L be 'the token begins a loop' and C be 'the token is a control-flow keyword other than return.'
- List L and C.
 - Find $P(L)$, $P(C)$, and $P(C^c)$.
- ☺ 7. *Two-neuron dropout mask.* A dropout layer has two neurons. A mask is a two-bit string, where 1 means active and 0 means dropped. All four masks are equally likely.
- Write Ω .
 - Define E: exactly one neuron is active, and N: no neuron is active.
 - Find $P(E)$, $P(N)$, and $P(E \cup N)$.
- ☺ 8. *Image-label audit.* One image is selected uniformly from $\Omega = \{\text{cat, dog, car, tree}\}$. Let A be 'animal,' V be 'vehicle,' and D be 'dog.'
- List A, V, and D.
 - Compute $P(A \cup V)$, $P(A \cap D)$, and $P((A \cup V)^c)$.
- ☺ 9. *Random permutation - algorithms.* A sorting algorithm receives six distinct elements in a uniformly random order.
- Find the probability that the largest element is in the first position.
 - Find the probability that the smallest element is in the last position.
 - Find the probability that the largest and smallest elements are adjacent.
 - Find the probability that the first two elements are already in increasing order.

! *Important problems*

- ! 10. *Three-unit-test report.* An AI-generated function is checked by three unit tests. Each test is pass (1) or fail (0), so an outcome is a three-bit string.
- Write Ω and state $|\Omega|$.
 - Let B be the event that at least two tests pass. List B.
 - Let C be the event that the first test fails. List $B \cap C$.
 - List B^c . If all outcomes are equally likely, compute $P(B)$.

- ! 11. *Chatbot exit reasons.* A chatbot session ends as success (0.56), timeout (0.18), safety block (0.12), invalid output (0.09), or server error (0.05).
- Verify that these values define a valid probability model.
 - The system retries after timeout, invalid output, or server error. Find the retry probability.
 - Find the probability that a session is not successful.
 - Are 'retry' and 'safety block' disjoint? Explain from the outcome definitions.
- ! 12. *Overlapping model-quality events.* In an audit of 1000 generated answers, 120 contain a hallucination H, 80 contain a citation error C, and 35 contain both.
- Estimate $P(H)$, $P(C)$, and $P(H \cap C)$ using relative frequencies.
 - Use inclusion-exclusion to estimate $P(H \cup C)$.
 - Estimate the probability that an answer contains neither problem.
 - How many answers contain a hallucination but no citation error?
- ! 13. *Shuffling three training shards.* Three training shards A, B, and C are processed in a uniformly random order.
- List the six outcomes in Ω .
 - Let E be 'A appears before B' and F be 'B appears before C.' List E, F, and $E \cap F$.
 - Compute $P(E \cup F)$.
- ! 14. *Hash collision.* Two distinct keys are each hashed into one of four buckets. An outcome is (i, j) , the bucket of the first and second key. Assume all 16 ordered pairs are equally likely.
- Describe Ω compactly and give $|\Omega|$.
 - List the collision event K.
 - Find $P(K)$ and $P(K^c)$.
 - Generalize the collision probability to m buckets.
- ! 15. *First successful compilation.* Let $K \in \{1, 2, 3, \dots\}$ be the attempt on which AI-generated code first compiles. A proposed model is $P(K = k) = 2^{-k}$ for $k = 1, 2, 3, \dots$
- Show that this is a valid probability model.
 - Find $P(K \text{ is odd})$.
 - Find $P(K > 3)$.
 - Write the event 'the code compiles within three attempts' and find its probability.
- ! 16. *Random binary string.* A binary string of length 8 is generated by choosing each bit independently with probability $1/2$.
- Find the probability that the string contains exactly four 1s.
 - Find the probability that it starts and ends with the same bit.
 - Find the probability that it contains no two consecutive 1s.

- ! 17. *Randomly selected program modules.* A software system contains 20 modules, of which 5 are security-critical. A testing procedure selects four distinct modules uniformly at random.
- Find the probability that exactly two selected modules are security-critical.
 - Find the probability that at least one selected module is security-critical.
 - Find the probability that none of the selected modules is security-critical.
- ! 18. *Countable sample space - number of attempts.* A computer repeatedly generates independent uniform random numbers from $\{1, 2, \dots, 10\}$ until 7 appears. Let X be the number of attempts required.
- State the sample space of X and decide whether it is finite or countably infinite.
 - Find $P(X = k)$ for $k = 1, 2, \dots$
 - Find $P(X > 5)$.
 - Find the probability that 7 appears for the first time on an even-numbered attempt.
- ! 19. *Infinite binary sequence.* A computer generates independent fair bits $X_1, X_2, \dots$. Let A be the event that the first occurrence of 1 is after position 10, and let T be the position of the first 1.
- Express A in terms of $X_1, \dots, X_{10}$ and find $P(A)$.
 - Find $P(T = n)$.
 - Find $P(T < \infty)$.

★ Challenging problems

- ★ 20. *Three-model ensemble.* Three binary classifiers output 0 or 1. Let $\Omega = \{0, 1\}^3$ and assume the eight triples are equally likely. Define M = 'majority vote is 1,' U = 'all three agree,' and $D = U^c$.
- List M , U , and D .
 - Find $M \cap D$ and $U \cup M$.
 - Compute the probabilities of these five events.
 - Explain why the uniform model is an assumption rather than a consequence of the sample space.

Hint. Enumerate the eight output triples first. Use $D = U^c$, then read each requested event directly from the list.

- ★ 21. *Selecting an evaluation subset.* A benchmark contains two sensitive items S_1, S_2 and three ordinary items O_1, O_2, O_3 . Two distinct items are chosen uniformly, without order.
- Describe the sample space and calculate its size.
 - Let A be 'both selected items are sensitive' and B be 'at least one is sensitive.' List A and calculate $|B|$.
 - Find $P(A)$, $P(B)$, and the probability that exactly one selected item is sensitive.
 - Recalculate $P(B)$ using a complement.

Hint. Treat outcomes as unordered two-item subsets. Count all choices with $C(5, 2)$, and use the complement 'both items are ordinary' for part (d).

- ★ 22. *A finite probability model for an AI pipeline.* An input is valid with probability 0.80 and invalid with probability 0.20. An invalid input stops immediately. Given a valid input, the model answer is accepted with probability 0.75 and rejected with probability 0.25.
- (a) Use the terminal paths I, VA, and VR as outcomes. Assign a probability to each path.
 - (b) Verify that the terminal-path probabilities form a valid model.
 - (c) Define S = 'pipeline succeeds' and R = 'human review is required,' where review follows invalid input or rejected output.
 - (d) Find $P(S)$, $P(R)$, and $P(S \cup R)$.

Hint. Use the three terminal paths I, VA, and VR. Multiply along each valid-input branch and check that the terminal-path probabilities sum to 1.

- ★ 23. *A countable model for repeated code generation.* Let $N \in \{1, 2, 3, \dots\}$ be the first attempt that produces syntactically valid code. Suppose $P(N = n) = c / 3^n$ for $n = 1, 2, 3, \dots$
- (a) Find c so that the probabilities sum to 1.
 - (b) Find the probability that the first valid program appears on an even-numbered attempt.
 - (c) Find the probability that the attempt number is a multiple of 3.
 - (d) Find $P(N > k)$ as a function of the positive integer k .

Hint. Normalize with a geometric series. The even and multiple-of-three cases are geometric subseries with ratios 1/9 and 1/27.

- ★ 24. *Random password.* A password consists of six distinct digits chosen from $\{0, 1, \dots, 9\}$. All ordered passwords are equally likely, and a leading zero is allowed.
- (a) Find the probability that the password contains exactly two even digits.
 - (b) Find the probability that the password starts with 0.
 - (c) Find the probability that the password contains at least one pair of consecutive digit values, such as 3 and 4, in any positions.

Hint. For part (a), choose the even and odd digits before arranging the six chosen digits. For part (c), note that the largest subset of $\{0, \dots, 9\}$ with no consecutive digits has size 5.

- ★ 25. *Hash-table collisions.* A hash function assigns each of five keys independently and uniformly to one of ten hash-table positions.
- (a) Find the probability that all keys are assigned to different positions.
 - (b) Find the probability that at least two keys collide.
 - (c) Find the probability that exactly one position contains two keys and every other occupied position contains one key.

Hint. Use 10^5 equally likely assignments. For exactly one double collision, choose the colliding key pair and its position, then place the other three keys injectively in the remaining positions.

- ★ 26. *Network packet transmission.* Each packet independently uses Route A, B, or C with probabilities $1/2$, $1/3$, and $1/6$, respectively. Five packets are sent.
- (a) Find the probability that exactly three packets use Route A.

- (b) Find the probability that at least one packet uses Route C.
- (c) Find the probability that exactly two different routes are used.
- (d) Find the probability that all three routes are used.

Hint. Use binomial or multinomial counting. For 'exactly two routes,' sum over the three route pairs and subtract the cases in which only one route from the pair is used.

- ★ 27. *Random search algorithm.* A search algorithm examines records independently. At each step, the desired record is found with probability p . Let X be the number of records examined until the first success.

- (a) Construct the probability model for X and find $P(X = n)$.
- (b) Find $P(X \leq n)$.
- (c) When $p = 0.2$, find the probability that more than 10 records are examined.
- (d) For $p = 0.2$, find the smallest n such that $P(X \leq n) \geq 0.95$.

Hint. This is a geometric model: $P(X = n) = (1-p)^{n-1}p$ and $P(X \leq n) = 1-(1-p)^n$. Use logarithms only in the final threshold step.

- ★ 28. *Randomized algorithm.* A randomized algorithm makes independent attempts, each succeeding with probability 0.3. It stops after its third success. Let X be the total number of attempts.

- (a) What values can X take?
- (b) Find $P(X = 3)$, $P(X = 4)$, and $P(X = 5)$.
- (c) Derive a formula for $P(X = n)$, $n \geq 3$.
- (d) Find $P(X > 10)$.
- (e) Explain why the sample space for X is countably infinite.

Hint. For $X = n$, the first $n-1$ attempts must contain exactly two successes and attempt n must succeed. For $X > 10$, count at most two successes among the first ten attempts.